

BEYOND GDP

Measuring Economic Health When Nature Counts

Green GDP, Planetary Boundaries, and the Doughnut of Social and Ecological Limits
July 2026

Sources: Zheng & Chen (2024), Frontiers in Environmental Science; Steffen et al. (2015), Science; Fanning & Raworth (2025), Nature

What GDP Fails to Count

GDP prices current production, not the future it borrows against.

Resource depletion is invisible

Felling a forest or overfishing raises GDP — with no debit for the depleted stock left behind.

Pollution is not a cost

Waste gas, sewage, and solid waste raise output, but their treatment cost never appears in the ledger.

Non-market welfare is excluded

Ecosystem services like clean air and water sit outside GDP, though they underpin all production.

"A country with vast forest reserves could boost its GDP by felling trees — causing significant environmental damage."

What Is Green GDP (GGDP)?

Definition

Green GDP subtracts the monetary cost of environmental damage from conventional GDP — both resource depletion and pollution.

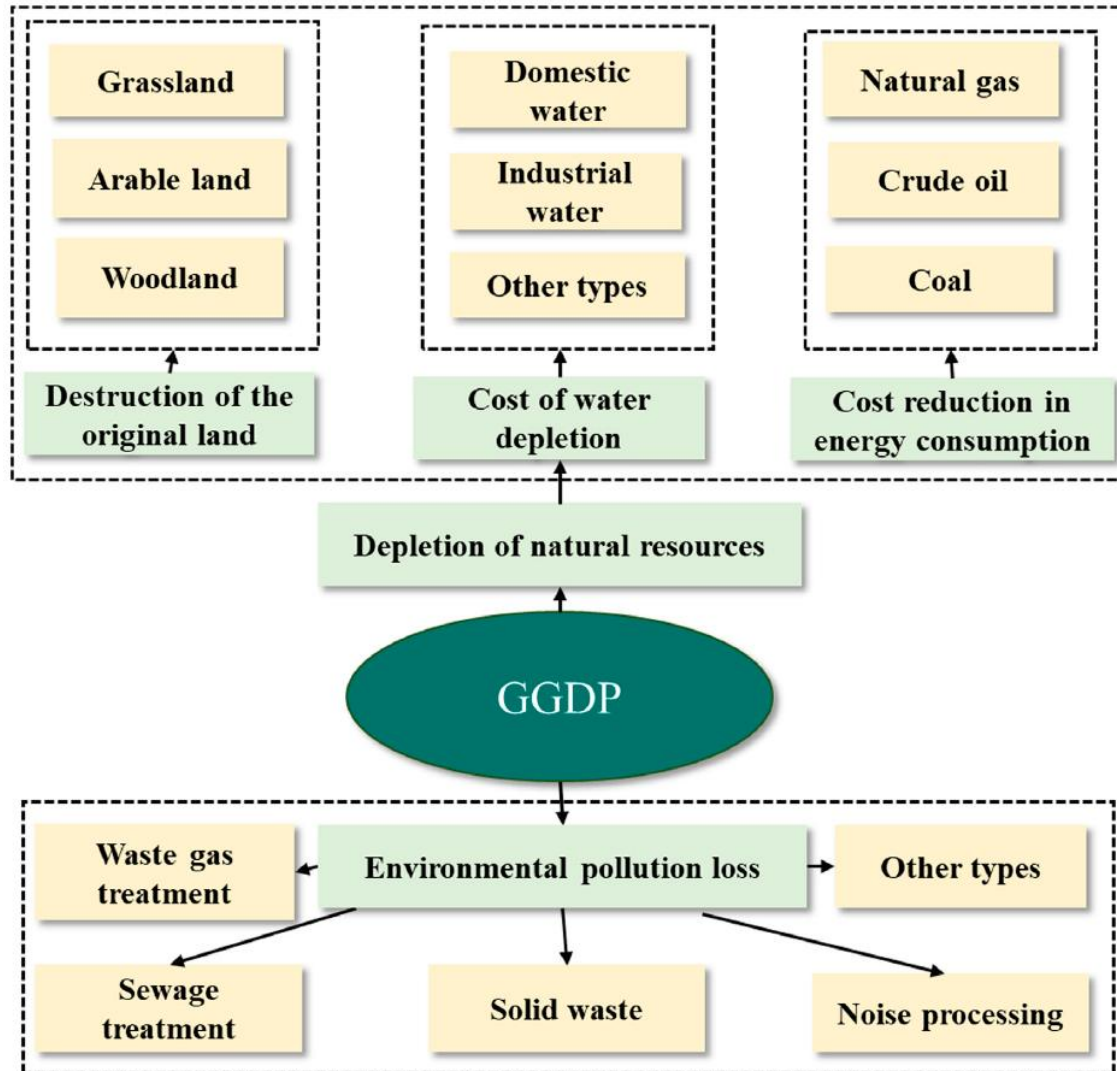
$$\text{GGDP} = \text{GDP} - (\text{R} + \text{E})$$

R = resource-depletion loss | E = pollution loss

Why it matters

- Reveals the true, sustainable size of an economy
- Penalizes growth that liquidates natural capital
- Gives one comparable metric for growth vs. conservation
- Tracks the GDP–GGDP 'gap' as a sustainability signal

The Structure of GGDP



Resource loss (R)

- Land destroyed (grass/arable/wood)
- Water consumed (domestic + industrial)
- Energy used (coal, oil, gas)

Pollution loss (E)

- Waste gas (esp. CO₂)
- Sewage treatment
- Solid waste disposal
- Noise & other pollution

Ecological Footprint & Biocapacity

Global hectare (gha)

- A hectare normalized to world-average biological productivity — a common currency across land types and oceans

Biocapacity

- The regenerative capacity of ecosystems — how much an area can renew each year, also in gha

Footprint (gha) = Quantity consumed ÷ Land yield, converted to global hectares

- 2025: humanity's Footprint is ~21.7 billion gha vs. ~12.2 billion gha of biocapacity — demand exceeds regeneration ~1.8×
- A country runs an ecological deficit when its Footprint exceeds its own biocapacity

This Paper's Model: 13 Sub-Criteria Under R + E

Resource Loss (R)

- Mineral & energy loss — coal, oil, gas consumption cost
- Land resource loss — grassland, arable land, woodland
- Water resource loss — domestic & industrial depletion

Pollution Loss (E)

- Sewage treatment (ST) cost
- Greenhouse gas damage — net CO₂ not reabsorbed
- Solid waste damage (SWD) — disposal/treatment cost

Source: Zheng & Chen (2024), §2.3, §3.2

14 secondary factors were screened via PCA down to 4 dominant drivers: energy consumption, CO₂ emissions, water & wastewater, and woodland/agricultural land use.

Resource Loss (R): The Underlying Formulas

Land destruction cost

= Σ (change in land area, by type) $\times$ (average market price of that land type, that year) — arable, woodland, and grassland computed separately.

Water depletion cost

= Σ (annual water consumption, by type) $\times$ (average unit price of that water type, that year) — domestic plus industrial.

Energy consumption cost

= Σ (annual consumption of coal, crude oil, natural gas) $\times$ (average price of each, that year).

These are all price $\times$ quantity formulas — GGDP accuracy hinges on the quality of the underlying quantity data. That's where measurement technology matters most.

Measuring Land Resource Loss: Surveys to Satellites

Traditional approach

- National yearbooks report land-use totals, often annually or less
- Land-use change detected via periodic surveys or censuses
- Prone to lag, self-reporting bias, coarse categories

Remote-sensing upgrade

- Landsat & Sentinel-2: land-cover classification at 10-30m, updated every few days
- Radar (SAR) sees through clouds — vital for tropical forest monitoring
- LiDAR estimates forest biomass directly, not just area
- Brazil's PRODES satellite system underpins \$3B+ in REDD+ payments

Measuring Water Resource Loss: The GRACE Revolution

Traditional approach

- Metered domestic & industrial withdrawal from utilities
- Groundwater tracked via sparse borehole well networks
- Blind to aquifer-level depletion beneath the surface

Remote-sensing upgrade

- GRACE / GRACE-FO satellites measure gravity-field changes to infer total terrestrial water storage
- Revealed major depletion in the Middle East, North China Plain, California, and NW India
- Combined with land-surface models, isolates the groundwater component
- Enables water-stress accounting even without dense well networks

Measuring Energy Consumption Loss

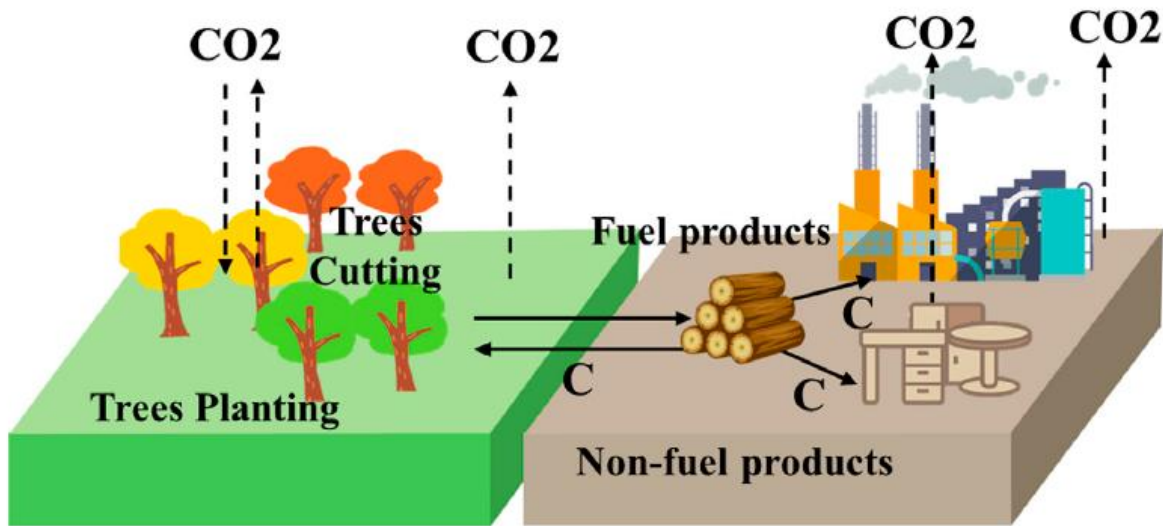
Traditional approach

- National energy balances from statistical agencies
- Annual/quarterly reporting — too coarse for short-term shocks
- National-level aggregation masks regional hotspots

Remote-sensing & modelling upgrade

- Nighttime-lights satellites (VIIRS/DMSP-OLS) proxy sub-national energy use nightly
- Satellite flaring detection tracks wasted energy at facility scale
- Smart-grid telemetry gives near-real-time consumption in advanced economies
- ML downscaling blends statistics with satellite proxies where data lags

Pollution Loss (E): The Carbon-Cycle Logic



Why net CO₂ matters

- CO₂ is the principal GHG indicator: long lifetime and a proxy for other pollutants
- Forests are a carbon sink — net emissions must subtract their uptake
- Cutting removes both the standing carbon stock and future sequestration

Converting Pollution to Dollars: The Monetary Step

Just like resource loss, every pollution indicator follows $\text{Cost (\$)} = \text{Quantity} \times \text{Price}$. The hard part is choosing a defensible price for something with no market.

$$\text{CO2 cost (\$)} = \text{Tons of CO2 emitted} \times \text{Carbon price (\$/ton)}$$

Carbon price sources: World Bank carbon-pricing metrics, EU ETS market price, or a government's official Social Cost of Carbon estimate

- Methane & other GHGs are first converted to CO2-equivalent using their Global Warming Potential (GWP) — e.g. methane $\approx 28\text{-}36\times$ CO2 impact over 100 years — then priced the same way
- Sewage & solid waste follow the identical logic: $\text{Cost} = \text{annual tonnes treated} \times \text{the unit treatment price charged that year}$
- The choice of price is the single biggest source of disagreement between different Green GDP estimates for the same country and year

Measuring CO2 & GHGs: New Instrumentation

OCO-2 / OCO-3 (NASA)

Satellites measuring atmospheric CO2 globally — detect regional sources and sinks at sub-city resolution.

Sentinel-5P (ESA)

Tracks methane, NO2, and trace gases daily — pinpoints industrial and farm emission hotspots.

Eddy covariance towers

Ground stations measuring real CO2 exchange between forest canopy and atmosphere.

Why this matters for GGDP

- Ecosystem process models (CASA, InVEST, Biome-BGC) combine satellite inputs with modeling to estimate flux where direct measurement is sparse
- This closes a major GGDP data gap: atmospheric concentration and forest-sink data can validate reported fuel-consumption statistics

Measuring Sewage, Solid Waste & Noise Pollution

Sewage / wastewater — cost = annual quantity treated × unit treatment price

Measurement upgrade: IoT-enabled treatment-plant sensors; satellite water-quality monitoring (chlorophyll, turbidity) for untreated discharge into rivers and coasts.

Solid waste — cost = annual quantity × unit disposal/treatment price

Measurement upgrade: satellite imagery tracking landfill footprint growth; RFID/weighing-based municipal collection data.

Noise — not separately priced in the paper's numerical model

Included conceptually under 'other pollution loss'. Urban acoustic sensor networks are emerging but remain far less standardized than carbon or water metrics.

Persistent Challenges in Measuring GGDP

Non-market valuation

Ecosystem services have no transaction to price — shadow-pricing them is inherently contestable.

No standardized global pricing

Carbon prices, water tariffs, and land values vary hugely by jurisdiction.

Historical data gaps

Consistent multi-decade series are rare outside a few well-studied countries.

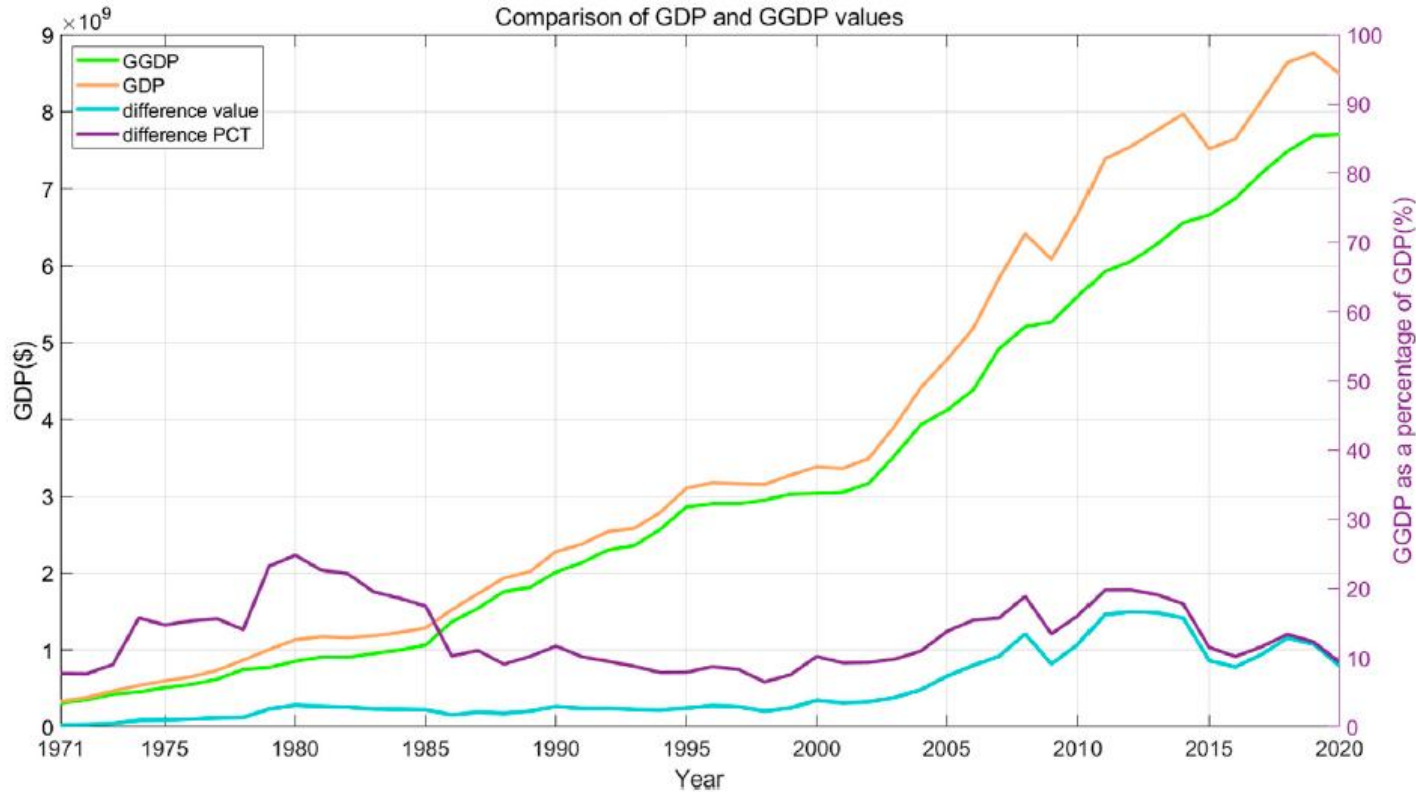
The informal economy

Illegal logging, artisanal mining, unregulated fishing go unrecorded even in physical terms.

Model & instrument uncertainty

Remote-sensing estimates carry their own error bounds that propagate into GGDP uncertainty.

GDP and Green GDP Are Diverging



Key finding

- Both GDP and GGDP grew continuously since 1970 — GGDP has grown markedly slower
- The widening gap reflects growth increasingly outpacing resource replenishment
- A declining GGDP/GDP ratio signals intensifying environmental exploitation

The Shrinking (But Still Dominant) Resource Share

90% → 80%

Share of global GDP from natural-resource-dependent activities, 1970s vs. 2020. A decline — but resource dependence still anchors most of global output.

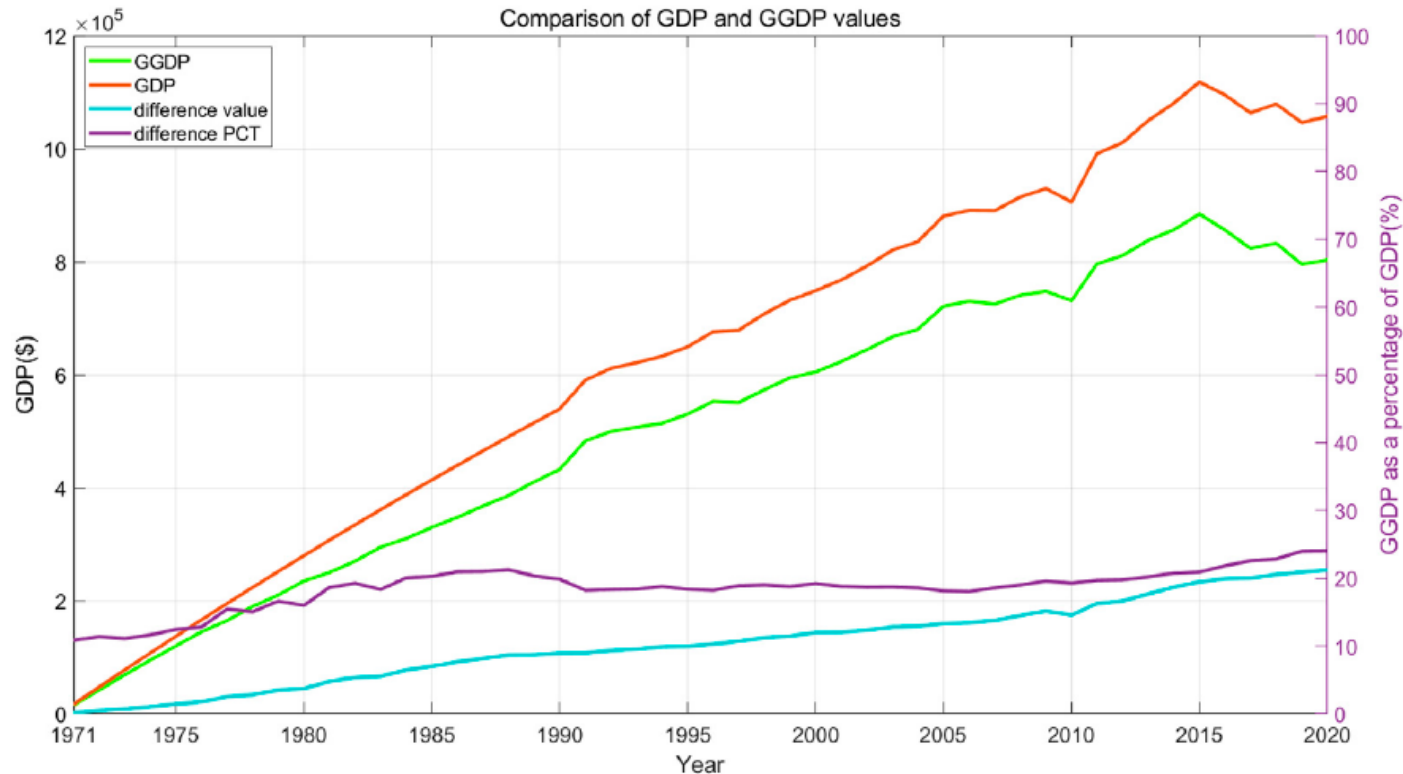
PCA: 4 dominant factors

Factor	Loading
Energy	0.93
CO2 emissions	0.979
Forest	0.495
Water	0.179

Energy and CO2 dominate GGDP variation; factors under 2% were excluded from the refined model.

Source: Zheng & Chen (2024), §3.2, Table 1

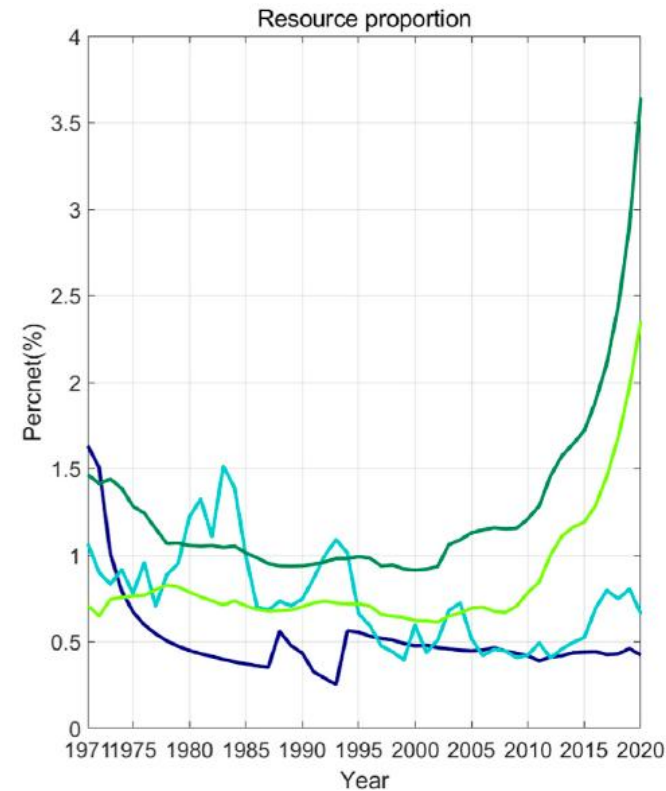
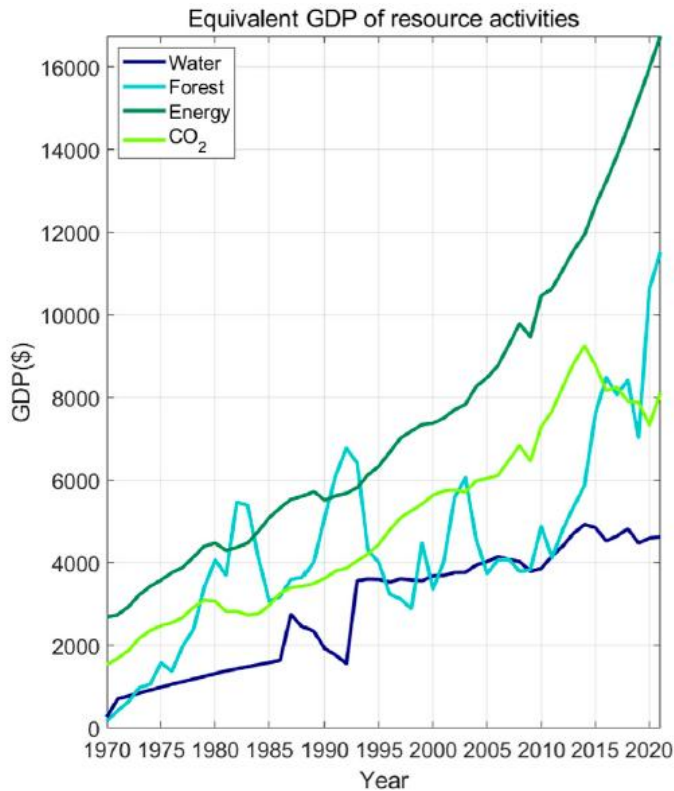
Brazil: GDP vs. Green GDP, 1971–2020



Reading the trend

- A declining GGDP/GDP ratio signals rising resource exploitation over time
- 2005–10: both GDP and GGDP fell alongside excessive forest use
- 2015–20: a smaller decline tied to over-exploited water & forest

Which Resources Drive Brazil's GGDP Gap?



Findings

- Energy shows the sharpest late-period rise, CO2 close behind
- Imported-resource value/GDP overtook environmental-resource value after 2006
- Coal, oil, and plant foods dominate long-run growth
- ESI & Improved ESI declined steadily 1970–2020

The Soybean–Deforestation–Amazon Tension

Soybean production

- A major GDP driver — Brazil is a leading global soybean producer/exporter
- But expansion drives deforestation, biodiversity loss, and higher GHG emissions

The Amazon rainforest

- Provides carbon sequestration, water-cycle regulation, and biodiversity
- Its degradation threatens both national and global wellbeing

The paper's recommendation: reduce dependence on monoculture farming and transition to techniques that restore natural ecosystems.

PART 6

The Planetary Boundaries Framework

If Green GDP prices what nature costs us, Planetary Boundaries define the limits within which any economic activity can safely operate.

Source: Steffen et al. (2015), 'Planetary boundaries: Guiding human development on a changing planet', Science 347(6223)

The Holocene, and a 'Safe Operating Space'

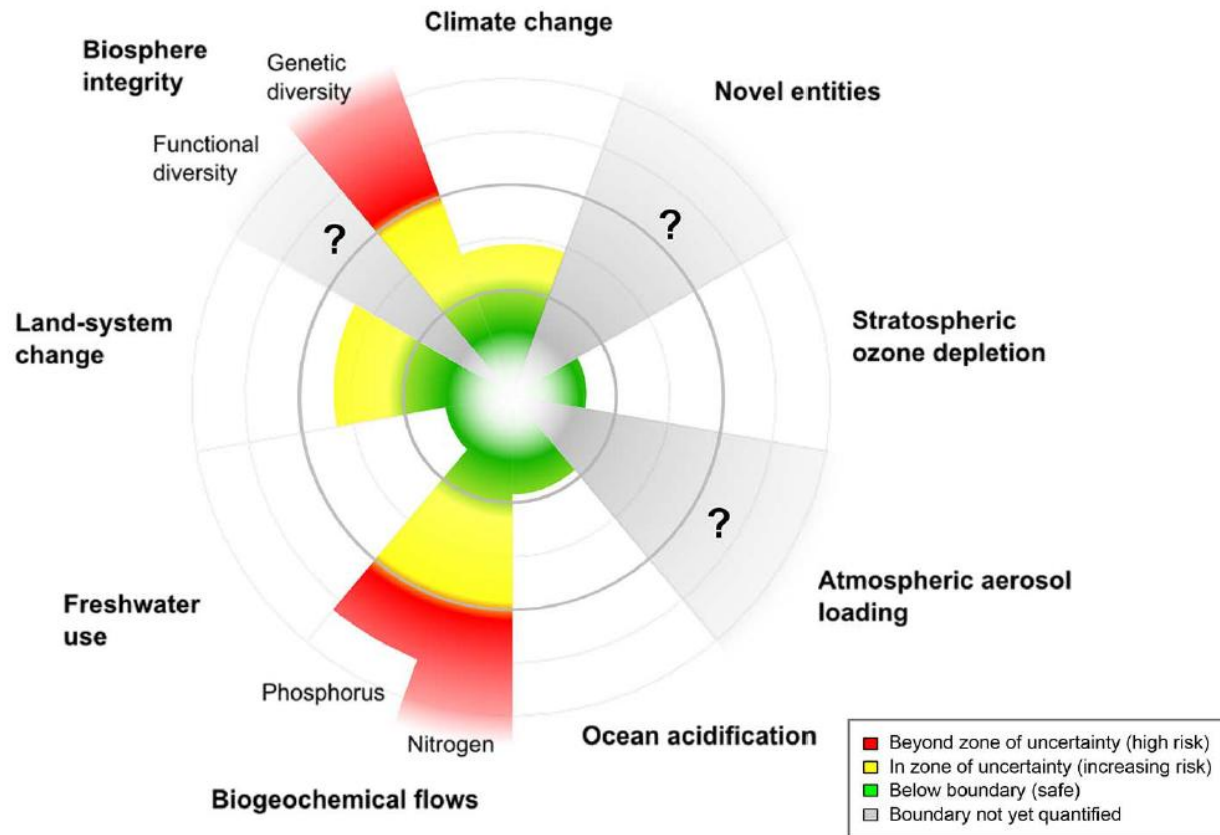
The geological argument

- The Holocene (last 11,700 years) is the only Earth state known to support complex civilization
- Its stability enabled agriculture, cities, and everything since
- Human activity now rivals geological forces — the proposed 'Anthropocene' epoch

The framework's logic

- A boundary is NOT the threshold — it sits upstream in a 'safe' zone, applying the precautionary principle
- Beyond it: rising risk; beyond the threshold: high odds of abrupt, possibly irreversible change
- 9 processes proposed in 2009

The Nine Boundaries: Current Status



Status at a glance

- Transgressed: climate change, biosphere integrity, biogeochemical flows (N & P), land-system change
- Regionally stressed: freshwater use, aerosol loading (South Asian monsoon)
- Within safe space: stratospheric ozone, ocean acidification
- Unquantified globally: aerosol loading, novel entities, biosphere function

Climate Change: The Geological Backstory

The process

- Climate is set by the balance of incoming solar and outgoing infrared energy
- Ice cores show CO₂ oscillated 180-280 ppm over 800,000 years, driven by orbital cycles
- The stable ~280 ppm Holocene allowed predictable seasons and agriculture
- Fossil fuels release carbon sequestered over millions of years, in a geological instant

Control variables & values

- Atmospheric CO₂ — boundary 350 ppm, current 398.5 ppm (>420 ppm today)
- Radiative forcing — boundary +1.0 W/m², current +2.3 W/m²
- Both were already transgressed as of this 2015 update
- Radiative forcing captures ALL anthropogenic drivers — generally the stricter test

Biosphere Integrity: Genetic & Functional Diversity

The process

- The biosphere regulates material/energy flows and buffers the Earth system against abrupt change
- Genetic diversity is the 'information bank' built over ~4 billion years of evolution
- Fossil record background extinction: ~1 species per million species-years
- Today's rate is estimated at 100-1000× that background rate

Control variables & values

- Extinction rate — boundary <10 E/MSY, current 100-1000 E/MSY
 - Extinctions per Million Species Years
- Biodiversity Intactness Index — boundary ≥90%, current 84% (S. Africa only)
- E/MSY = extinctions per million species-years, an interim variable
- One of two 'core' boundaries — its transgression alone could shift Earth-system state

Biogeochemical Flows: Nitrogen & Phosphorus

The process

- For most of Earth's history, nitrogen entered ecosystems only via slow biological fixation
- The Haber-Bosch process (1909) now fixes more N than all natural terrestrial processes combined
- Phosphorus cycles over geological time, released only by rock weathering — mined fertilizer is one-way
- Excess N & P runoff drives eutrophication and aquatic 'dead zones'

Control variables & values

- P: freshwater→ocean flow — boundary 11 Tg/yr, current ~22 Tg/yr
- P: fertilizer→soils (regional) — boundary 6.2 Tg/yr, current ~14 Tg/yr
- N: fixation — boundary 62 Tg/yr, current ~150 Tg/yr
- Both flows exceed double their boundary — among the most severely transgressed

Land-System Change: Forests as Climate Regulators

The process

- Forests regulate climate directly via energy, water, and momentum exchange, not just carbon storage
- Tropical forests cool via evapotranspiration — clearing removes this effect regionally and globally
- Boreal forests affect surface albedo — dark canopy absorbs more solar energy than snow
- The control variable shifted from cropland area to forest-cover-remaining, tracked per biome

Control variables & values

- Global boundary: 75% of original forest cover (current: 62%)
- Tropical boundary: 85% of potential forest
- Temperate boundary: 50% of potential forest
- Boreal boundary: 85% of potential forest

Freshwater Use: Environmental Water Flows

The process

- Freshwater ecosystems depend on natural flow variability, not just total annual volume
- 'Consumptive blue water use' captures water permanently removed from the cycle
- The Environmental Water Flow (EWF) concept defines flow needed for a fair-to-good ecosystem state
- Excess withdrawal risks wetland collapse, saltwater intrusion, and fish population crashes

Control variables & values

- Global blue water use — boundary 4000 km³/yr, current ~2600 km³/yr
- Basin low-flow withdrawal limit — 25% of mean monthly flow
- Globally safe, but this masks severe regional crises (Middle East, NW India, N. China)
- Now well-documented via GRACE satellite gravimetry

Ozone & Ocean Acidification

Stratospheric ozone depletion

- The ozone layer formed ~600 million years ago as oxygen accumulated — it shields land from UV and enabled life to colonize land
- CFCs catalytically destroy ozone molecules high above the surface
- Only transgressed over Antarctica in austral spring (~200 DU vs. 275 DU boundary)
- Recovering after the Montreal Protocol phased out CFCs — a rare success story

Ocean acidification

- Oceans absorb ~30% of anthropogenic CO₂, forming carbonic acid that lowers pH
- This reduces carbonate-ion concentration, undermining shell/skeleton-building organisms like coral and shellfish
- ~84% of pre-industrial aragonite saturation remains — within the safe zone
- Would stay safe if the 350 ppm CO₂ boundary were respected

Aerosol Loading & Novel Entities

Atmospheric aerosol loading

- Black carbon, sulfates, and nitrates from combustion scatter/absorb sunlight
- Alters regional atmospheric circulation — can weaken monsoons billions depend on
- Only regionally quantified (South Asian monsoon)
- Current AOD ~0.3, within the 0.25-0.50 zone of uncertainty

Novel entities

- Synthetic chemicals, plastics, and modified organisms with no evolutionary precedent
- Over 100,000 substances now in global commerce
- The CFC/ozone case is the cautionary template: risks discovered only after global-scale damage
- No global control variable yet defined — an unquantified, growing risk

All Nine Boundaries: Control Variables (1 of 2)

Process	Boundary	Current value
Climate change (CO ₂ / forcing)	350 ppm / +1.0 W/m ²	398.5 ppm / +2.3 W/m ²
Biosphere integrity	<10 E/MSY / ≥90% BII	100–1000 E/MSY / 84%*
Biogeochemical flows (P / N)	11 Tg P; 62 Tg N per yr	~22 Tg P; ~150 Tg N per yr
Land-system change	75% of original forest cover	62%
Freshwater use	4000 km ³ /yr	~2600 km ³ /yr

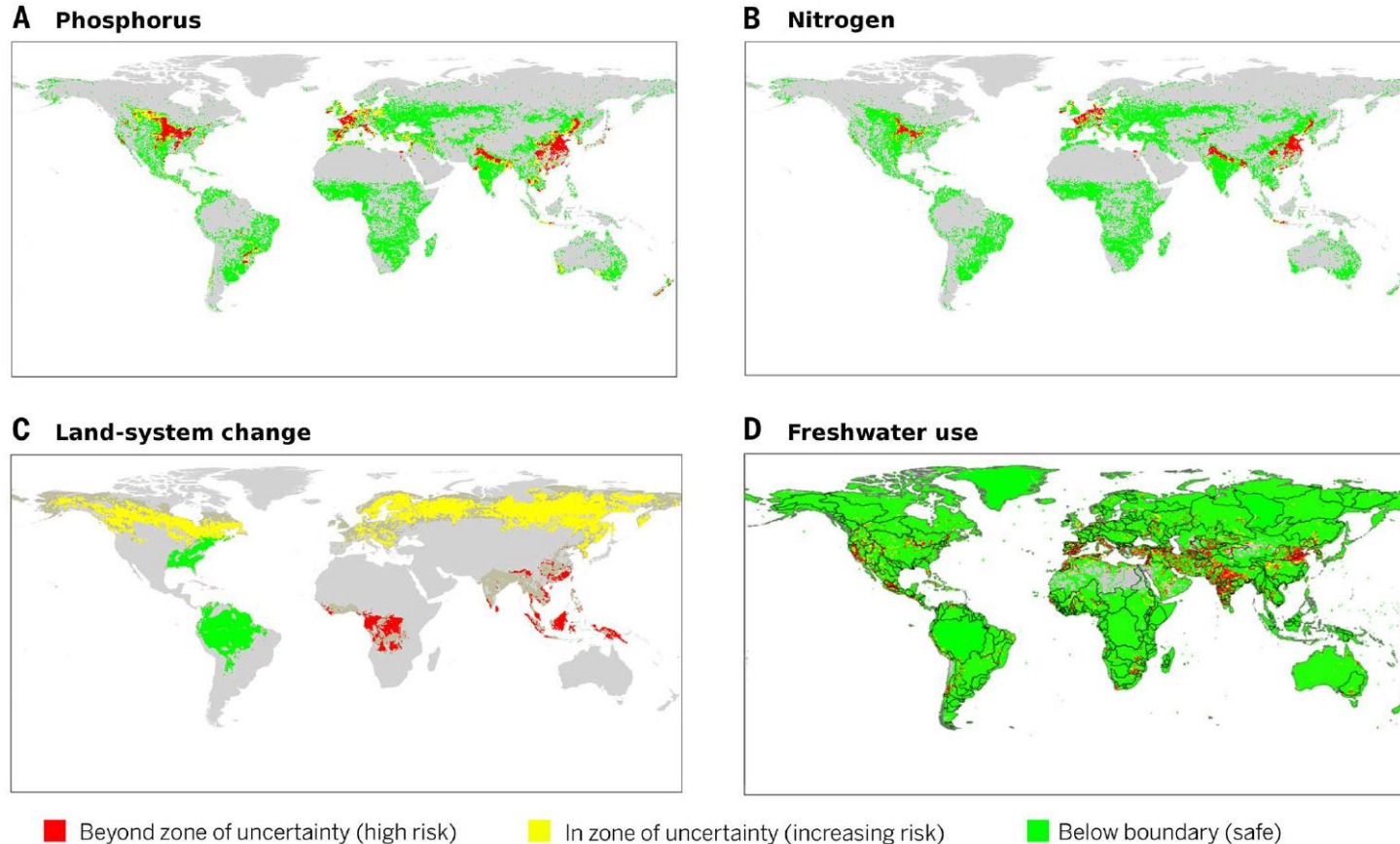
*BII measured for southern Africa only. Source: Steffen et al. (2015), Table 1

All Nine Boundaries: Control Variables (2 of 2)

Process	Boundary	Current value
Ocean acidification	≥80% of pre-industrial saturation	~84% of pre-industrial
Stratospheric ozone	<5% reduction (275 DU)	~200 DU (Antarctic spring only)
Atmospheric aerosol loading	0.25 AOD (S. Asia, regional)	~0.30 AOD
Novel entities	No control variable defined	—

Source: Steffen et al. (2015), Table 1

Global Averages Hide Regional Crises



Why regional matters

- A 'safe' global average can hide severely transgressed hotspots
- Freshwater withdrawal matters most at the river-basin scale
- Land-system change works through specific forest biomes
- This is why 2015 introduced two-tier (global + regional) boundaries

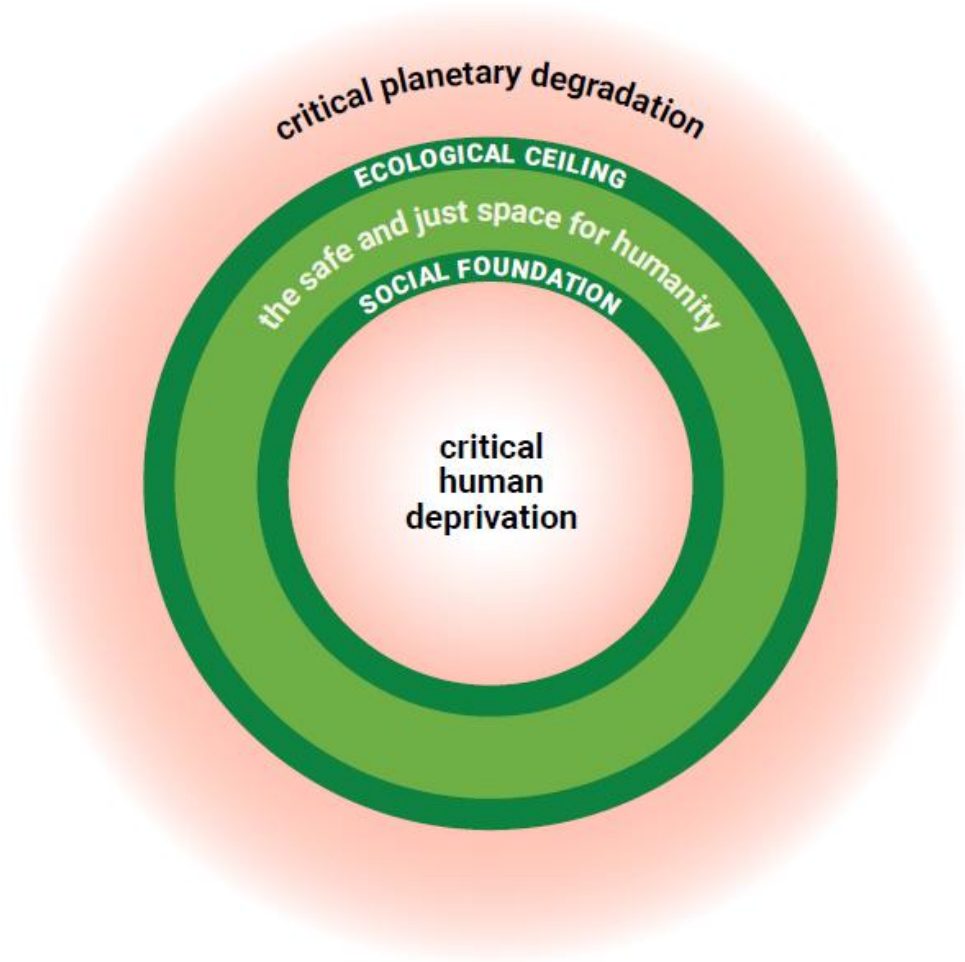
PART 7

The Doughnut: Social Foundation Meets Ecological Ceiling

Planetary boundaries define the outer limit. But an economy that stays inside them while leaving billions in need isn't a success either — the Doughnut holds both truths at once.

Source: Fanning, A.L. & Raworth, K. (2025), 'Doughnut of social and planetary boundaries monitors a world out of balance', Nature 646:47-56

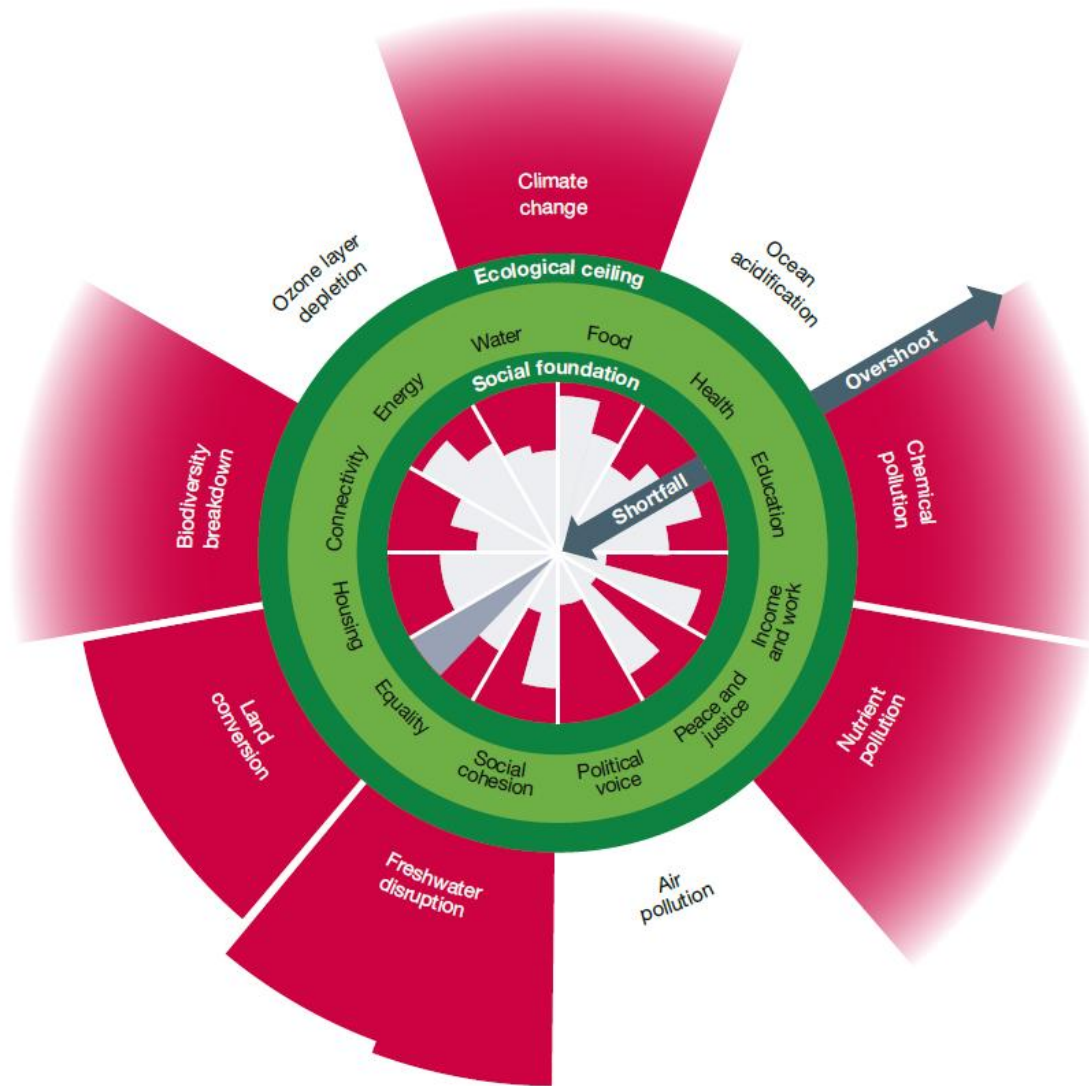
What Is the Doughnut?



Two concentric rings

- Inner ring — the social foundation: below it lies critical human deprivation (hunger, no clean water, no political voice)
- Outer ring — the ecological ceiling: beyond it lies critical planetary degradation (a transgressed planetary boundary)
- The doughnut-shaped space between them is the 21st-century goal: meeting everyone's needs within the means of the living planet

The 21 Dimensions, 2000–2022



12 social + 9 ecological

- Social: food, health, education, income & work, water, energy, connectivity, housing, equality, social cohesion, political voice, peace & justice
- Ecological: the same nine planetary boundaries from Part 6, renamed for accessibility (e.g. 'biodiversity breakdown')
- 35 indicators in total measure these 21 dimensions

Current Global Status, 2022

35%

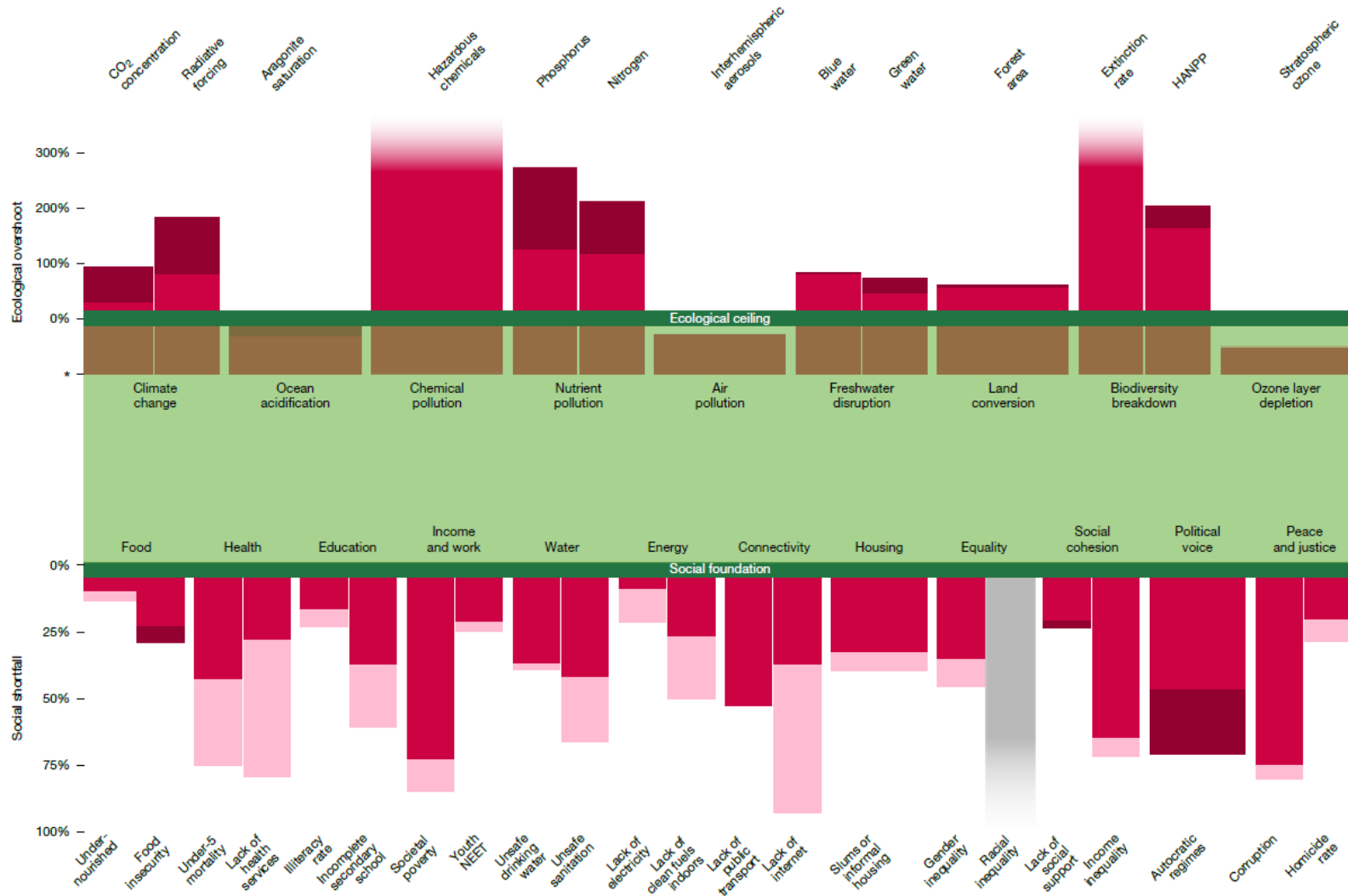
median social shortfall — around 3 billion people below the minimum standard on a typical indicator

96%

median ecological overshoot — nearly double the safe boundary, across the 9 planetary limits

- Social range: from 9% lacking electricity access to 75% perceiving widespread corruption
- Ecological range: from ~50% below the ozone boundary to over 10× beyond the safe limits for chemical pollution and species extinction
- Humanity now overshoots at least six of the nine planetary boundaries

A Bar Chart, Because Wedges Can Mislead



Why add a bar chart?

- Wedge length scales area quadratically — small changes can look far more dramatic than they are
- Bar charts avoid that distortion, showing the true linear change
- Light red = improvement; dark red = worsening; grey = missing data

Social Progress: Modest, and Too Slow

What improved (2000–2022)

- Indicators with most of humanity in shortfall halved: from 10 down to 5
- 13 of 21 indicators show discernible reductions in shortfall
- Biggest gains: internet, health services, child survival, sanitation, clean fuels — 24-56% escaped shortfall

What didn't

- 5 indicators barely changed: undernourishment, youth unemployment, unsafe water, social support, corruption perception
- 2 indicators worsened: food insecurity and autocratic-regime prevalence
- To hit the 2030 social goal, improvement must accelerate $\sim 4.9\times$ on median

Ecological Overshoot: Rapid and Worsening

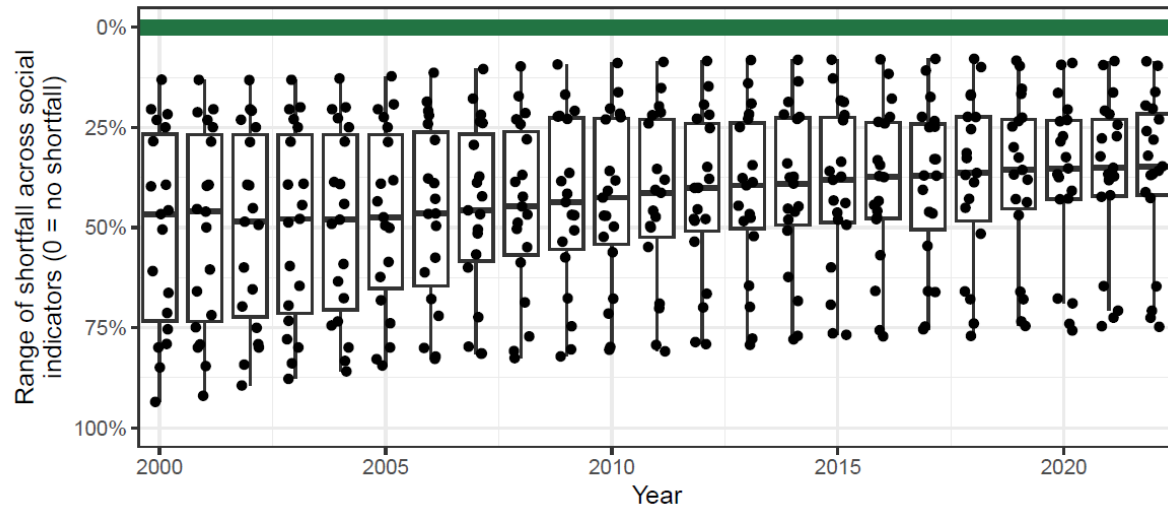
The trend since 2000

- 9 of 10 time-series ecological indicators worsened significantly (only ozone stayed roughly flat)
- 2 indicators worsened 70-80%: nitrogen use, green-water disruption
- 4 indicators more than doubled overshoot: CO₂, radiative forcing, hazardous chemicals, phosphorus

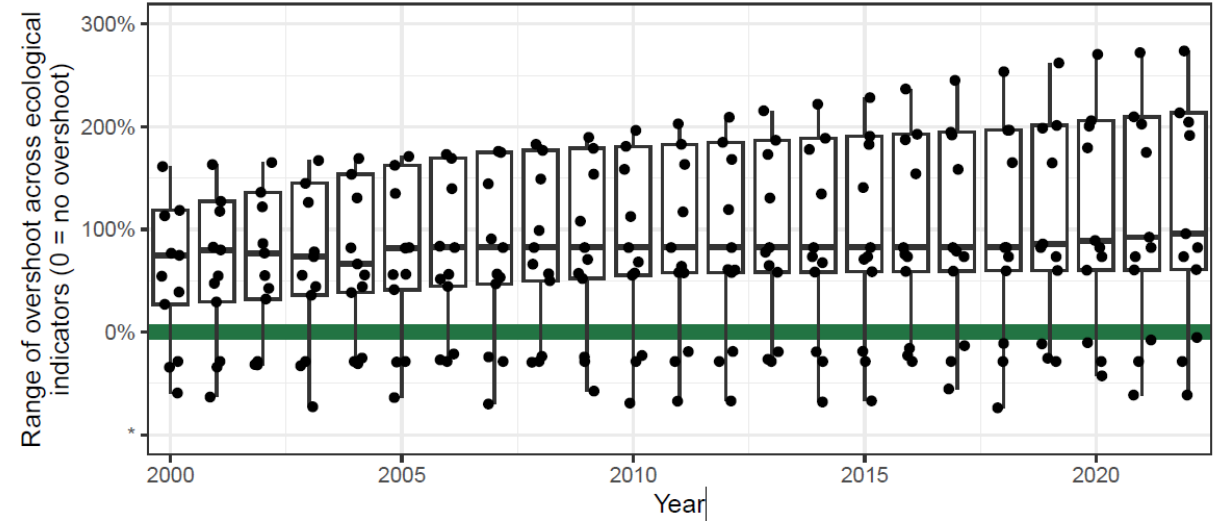
What it would take to fix by 2050

- Median overshoot rose from 75% (2000) to 96% (2022) of the ecological ceiling
- Degradation must not just stop, but reverse at ~6.9 percentage points per year on median
- The paper calls this pathway 'very low probability' given historical trends

Watching the Gap Widen, Year by Year



Range of social shortfall across indicators narrows from 47% (2000) to 35% (2022) median, IQR 27-73% to 22-42%



Range of ecological overshoot across indicators widens from 75% (2000) to 96% (2022) median, IQR 27-119% to 61-213%

Disaggregating by Income: Three Country Clusters

Global averages hide who is actually causing overshoot, and who is actually experiencing shortfall.

Poorest 40% of countries

42% of world population. Least ecological overshoot — but the greatest social shortfall across 19 of 21 indicators.

Middle 40% of countries

43% of world population. Contributions to overshoot and shortfall are both roughly proportional to population share.

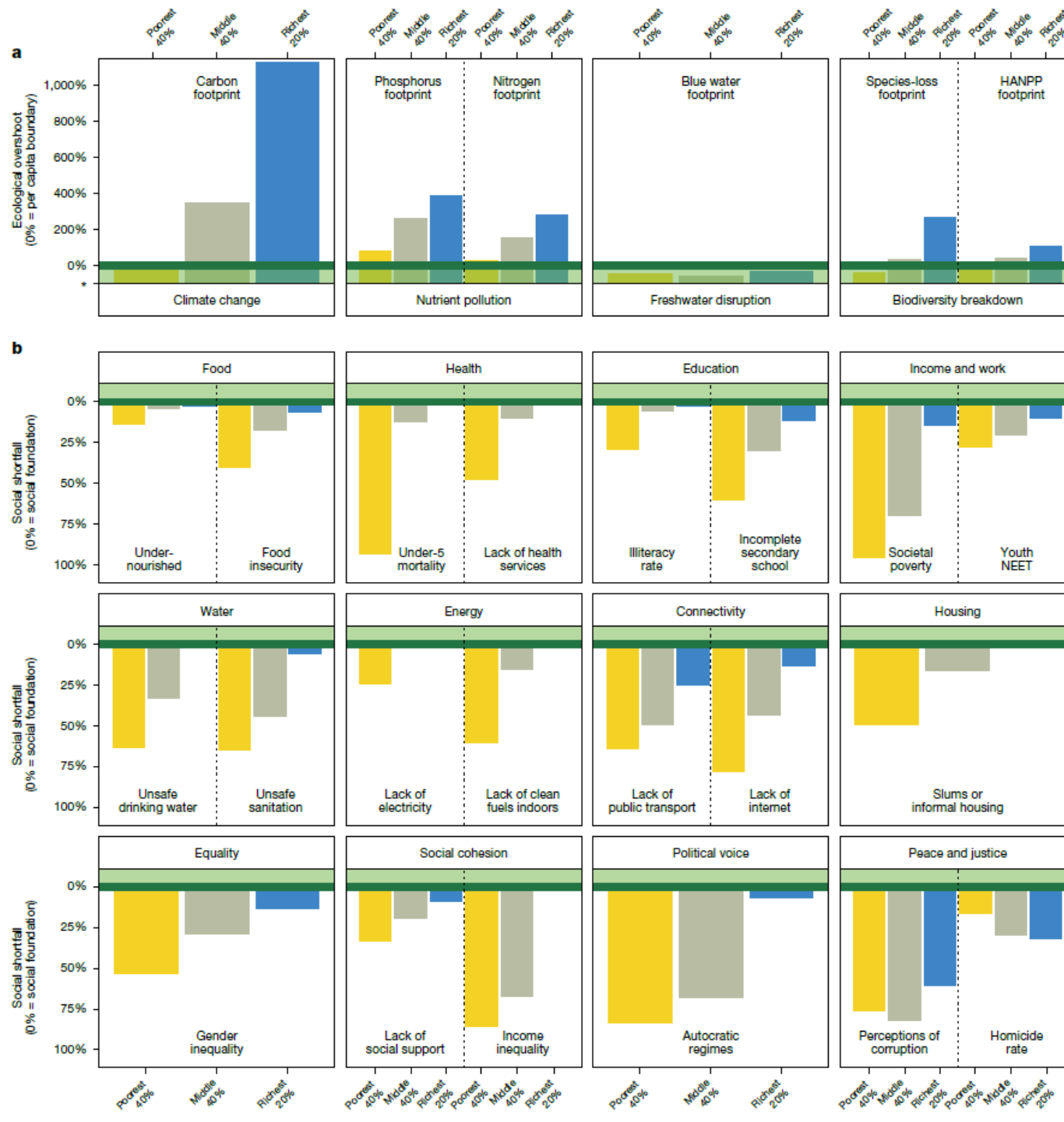
Richest 20% of countries

Only 15% of world population. Least social shortfall — but the greatest ecological impact of any cluster, by a wide margin.

Ecological Overshoot by Country Cluster

The scale of inequality

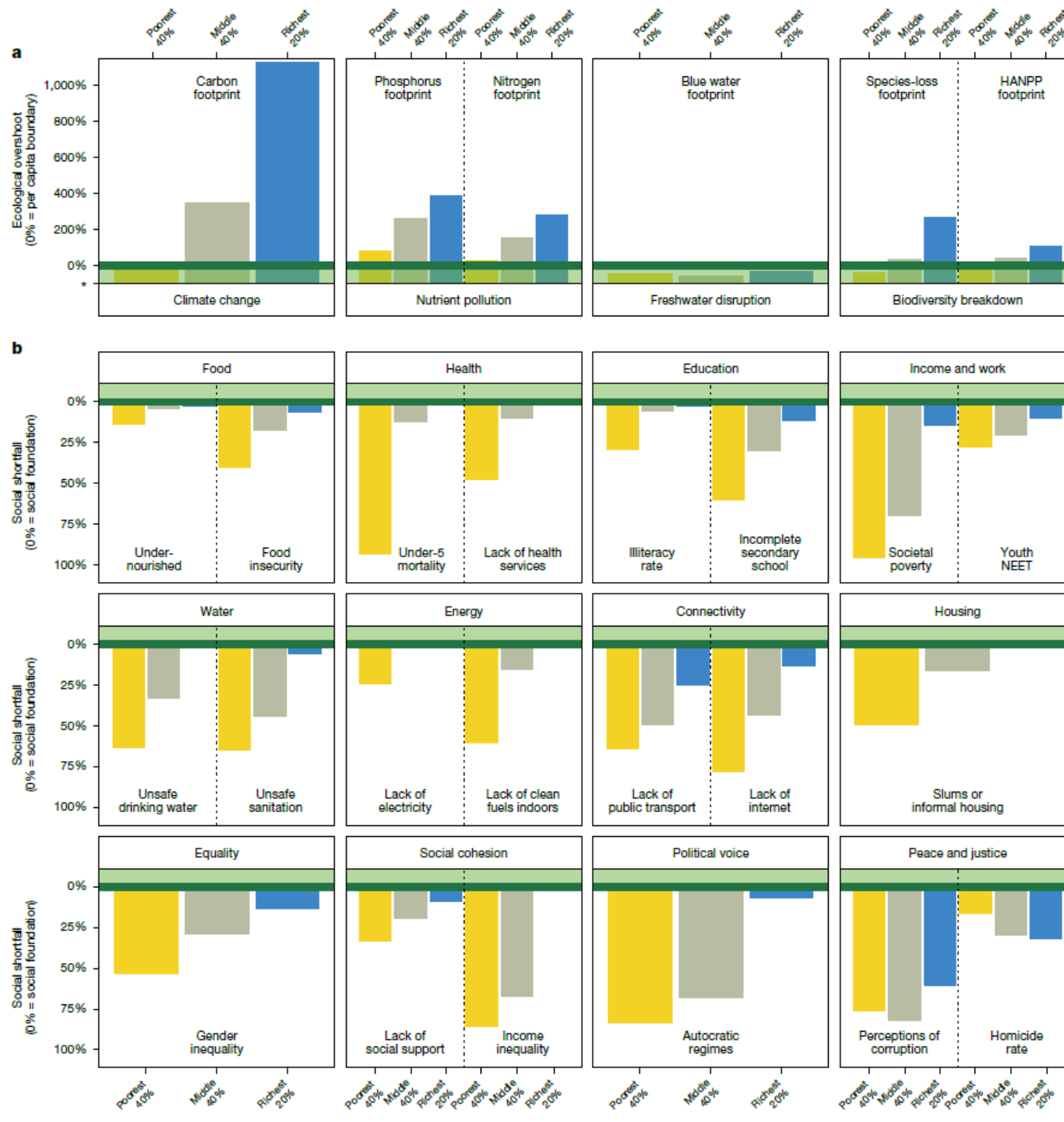
- The richest 20% and middle 40% both overshoot 5 of 6 per capita boundaries
- Median overshoot is nearly 3× larger in the richest 20% than the middle 40% (273% vs. 96%)
- The poorest 40% stays within nearly all per capita boundaries (median: just 1% over)



Social Shortfall by Country Cluster

The scale of deprivation

- Poorest 40% shortfall is worst for physical necessities: health, energy, housing
- Median shortfall share is double the middle 40% and 9× the richest 20% (60% vs. 29% vs. 6.6%)
- Exceptions: corruption perception is worst in the middle 40%; homicide rate is worst in the richest 20% (driven by 2 outlier countries)



Who Actually Causes the Damage?

**15% of people,
26-73% of overshoot**

The richest 20% of countries hold just 15% of world population but drive a disproportionate share (median 44%) of global ecological overshoot

**42% of people,
27-97% of shortfall**

The poorest 40% of countries hold 42% of world population, contribute almost nothing to overshoot (median 4%), yet hold most of the social shortfall (median 63%)

Inequality Within Countries?

If splitting the world into rich, middle, and poor countries reveals this much inequality — the same logic applied within a single country would reveal even more.

- The paper disaggregates nations into income clusters — the same consumption-based footprint method could disaggregate a country's own population into income deciles
- A country's richest 1% or 10% likely overshoot per-capita planetary boundaries by many multiples of its poorest 50% — mirroring the richest-20%-nations pattern shown here
- A domestic doughnut could reveal that a country judged 'sustainable' on average is actually two very different populations: one deep in ecological overshoot, one still short of the social foundation